METHODOLOGY — draft 2026-08-28

Status: complete-details draft for the author to rewrite. Every concrete value carries a pointer (`file:line` in the repo at HEAD `8b29524`, branch `4dof-cartesian`; artifact paths under the off-cluster mirror `~/workspace/final_rr_artifacts_2026-08-24/` = `M/`; cluster-only source under `/cluster/tufts/shortlab/jstale02/` = `C/`). Where sources disagree both values are given and flagged **[CHECK]**; the full **[CHECK]** list is collected in §8. Precedence when documents conflict: `paper/AUDIT_results_2026-08-28.md` > `paper/PREREG_final_round_robin_2026-08-23.md` (+A1–A10) > code > older narrative docs. Result numbers are deliberately *not* reproduced here except where they characterise an instrument (fidelity, survivors, ceilings); results live in `paper/RESULTS_TABLE_2026-08-25.md` (regenerate with `analysis/results_table.py`).

Notation: `d{source}_{learner}[-dense]_s{seed}`; sources dH (human), dDP (diffusion-policy teacher), dR2D (r2dreamer teacher); learners DP, RLPD, r2dreamer (r2d), dv3.

> **LIVING UPDATES (2026-08-29, kept current by the agent during the cluster block).** Sections carry an
>
> `UPDATE 08-29` block wherever the recipe, seeds, or rules changed after this draft. Precedence: PREREG amendments
>
> A16–A19 > these blocks > the 08-28 text. Stable material the user is writing lives outside this file.

1. Real robot and demonstration data

1.1 Platform and teleoperation

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item value pointer
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Robot Kinova gen3-lite, 6 revolute arm joints (joint_1..6) + trial_reader.py:8-9,23-25; kinova.py:1-8; URDF
2-finger gen3-lite gripper gen3_lite_2f_robotiq_85.urdf (misnomer: it is the kortex
(kortex_description/grippers/gen3_lite_2f); ROS namespace gen3-lite 2F gripper, not a Robotiq 2F-85 —
/my_gen3_lite paper/real2sim_follower_lab_2026-08-23.md:239)
Teleop modality Cartesian-velocity joystick teleop; the operator commanded wy\ ), tool setpoint clamped to a
base-frame workspace box baselines/cartesian_env.py:3-6,49,56,80-84;
end-effector velocity in [x, y, z, pitch] (roll/yaw fixed), WS_LO=(0.30,-0.25,0.015), WS_HI=(0.80,0.25,0.60)
baselines/genesis_can_env.py:31-41
linear cap 0.11 m/s (VCAP), pitch-rate cap 1.0 rad/s
(PITCH_CAP, = measured max \
4-DOF verification recorded wx=wz=0 in 25/25 inspected demos, wy≠0 in 25/25; v\ = [0.11, 0.11, 0.11, 0, 1.0, 0];
the plugin's Jacobian cartesian_env.py:80-84; measured from
over all 96 *_cartesian.npy tapes per-dim max \ controller left roll/yaw free (drift up to 1.03/1.34 rad)
inthewild_trials/*_cartesian.npy
Gripper signal base_feedback…gripper_feedback[0].motor[0].position, 0 open trial_reader.py:14; example.py:206-
207;
… 100 closed (observed range −0.98…100.69 in tapes); can_pos_recovery/replay_harness.py:60
"closed" threshold GP_CLOSE = 30
```

1.2 What was recorded (ROS bags → `inthewild_trials/`)

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item value pointer
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Topics (original reader) /my_gen3_lite/joint_states (position[:6]), /joy (read, never trial_reader.py:1-2,11-
26,30,50-51
stored), /my_gen3_lite/base_feedback (tool_pose
x,y,z,θx,θy,θz; gripper motor position)
Topics (cartesian reader) additionally /my_gen3_lite/in/cartesian_velocity trial_reader_cartesian.py:17,30-32,47-
56
(TwistCommand) and base_feedback.tool_twist_*;
/reward,/success deliberately not extracted (untrusted auto-
detector)
Windowing frame emitted whenever ≥1/60 s elapsed and both topics have trial_reader.py:32,43-54
≥1 msg in the window; values are per-window means; last
frame dropped
Effective rate nominal "30 Hz" (DATA_HZ = 30 # approx -- timestamps weren't example.py:276;
CAN_STARTING_POSITION.md:188;
saved; user_232 was ~31.6 Hz (919 frames / 29.06 s)); cartesian_env.py:51 (DT = 0.025)
joint_states itself is ~40 Hz (dt p95 25.4 ms);
*_cartesian.npy for uid 232 has 1159 frames (~39.9 Hz)
[CHECK 1]
Output (<uid>_episodes.npy) pickled dict {'vel_cmd': (T,6) float64, 'gripper_pos': trial_reader.py:69-74;
example.py:284;
(T,1)}. NB: despite the key name, vel_cmd is the measured real2sim_follower_lab_2026-08-23.md:85-86;
joint position stream (/joint_states position[:6]), replayed trial_reader_cartesian.py:11
as PD position targets; several docs call it "commanded
joint targets" [CHECK 2]
Superset (<uid>_cartesian.npy) joint_pos, vel_cmd(=joint_pos), tool_pose (m; θ in degrees),
trial_reader_cartesian.py:10-16,79-80;
tool_twist, gripper_pos, cartesian_velocity, can_pos_recovery/read_bag.py:7
cartvel_reference_frame float32; tool_pose is in the ROBOT
BASE frame; sim world z = base z + 0.05
Cameras per-trial MP4s (raw/user_<id>/cam_dev_video0 bottom-up CAN_STARTING_POSITION.md:123-124,189-190,218,283-
284;
through the translucent shelf, cam_dev_video4 top view of HANDOFF_PLAN.md:80-82
the start), stored rotated ~90°, no factory calibration;
resolution/model not documented anywhere [CHECK 3]; cameras
are NOT in the bag reader and NOT used by any learner (sim
cameras only, §3.2)
Object pose never recorded (motivates §2.3) CAN_STARTING_POSITION.md:8-9
Requirements original reader needs ROS1 rosbag; cartesian reader uses trial_reader.py:1;
trial_reader_cartesian.py:5-7,38
pure-python rosbags; raw bags are not on the devbox
```

1.3 Trial census and labels

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96 *_episodes.npy files on disk, uids 232–335 (missing ls inthewild_trials; trial_reader.py:78-86; kinova.py:16-
21
241,264,271,272,291,292,323,332); = union of the label lists
in kinova.py
93 trials in can_pos_recovery/trial_placements.json and PAPER_PLAN.md:75-80; paper/METHODOLOGY.md:65-72
baselines/demo_manifest.json (77 success / 16 fail). uids
253, 285, 289 (fail-list) are on disk but absent from both,
reason undocumented [CHECK 4]. Docs say "93 in-the-wild
trials recorded"
2 stubs 290, 322: gripper never closes; excluded CLAUDE.md; CAN_STARTING_POSITION.md:56,68,72,214
91 graded = baselines/demo_manifest_auth.json (75 success / 16 measured
fail)
16 fail-labeled 238 240 249 260 268 270 282 288 296 307 310 312 313 314 324 demo_manifest.json;
CAN_STARTING_POSITION.md:154,211
334 (label bug: 322 and 331 appear in both a success and a
fail list in kinova.py)
9 success-labeled but never pick under replay: 236 237 239 244 demo_manifest_auth.json
255 261 294 303 331
66 IL-usable human demos = 75 − 9; by furthest replay stage: PAPER_PLAN.md:60-71,75-80; crosstab of
picked-only 2 / placed 38 / contact 6 / nested 20 demo_manifest_auth.json
Frames 66 IL tapes = 132,365 frames; all 96 tapes min 356 / median measured from n fields
1530 / max 6233
Rule IL learners never train on failed demos (user rule PAPER_PLAN.md:357; METHODOLOGY.md:154-156
2026-07-30); fails are RL/WM negative data only
`-p 0 1 2` can-position bucket filter in example.py (nominal marked example.py:9,17,102-104;
```

```
circles POSITION_0=(0.4381,0.1), _1=(0.4381,-0.05), CAN_STARTING_POSITION.md:11-12,32-36
_2=(0.4381,-0.2); buckets wrong by up to ~5 cm, superseded
by per-trial recovery §2.3); bucket counts 38/27/28
Unlabeled 200–231 never ingested ("~130 recordings" claim inconsistent with a HANDOFF_PLAN.md:91-93
32-wide range [CHECK 5])
```

1.4 Task and geometry

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item value pointer
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Task pick a soup can from the table region and place/nest it paper/METHODOLOGY.md:45-48; PREREG §0
against a static goal can on a shelf; stages picked → placed
→ contact → nested; the study is scoped to the pick phase
Can (real) Campbell's soup can 66 × 101 mm, ≈0.35 kg CAN_STARTING_POSITION.md:170-171
Can (sim) cylinder r 0.033, h 0.101, ρ 1000 → 0.345 kg; can_friction trial_placements.json['world'];
replay_harness.py:125-128;
0.2 (inert: finger–can pair takes max(1.0,0.2)) gripper_lab_2026-08-25.md:62-64
Goal can same cylinder, ρ 1000, friction 2.0, static at replay_harness.py:51,129-132
STATIC_BOTTLE_POSITION = (0.672, −0.221, 0.19)
Goal recovery user-validated demos 233 + 242 ("pick/place/slide perfect") replay_harness.py:43-51;
must end touching the goal → two-circle intersection, south can_pos_recovery/goal_from_slides.py:1-19,
candidate chosen (matches slide directions). Rejected goal_from_slides.json; CAN_STARTING_POSITION.md:450-476;
alternatives: fixed-radius ring fit on 61/75 real tool_pose CLAUDE.md 07-20
slides → (0.662, −0.057) (rms 25 mm; biased by one-sided
north approaches); placement-cluster median (0.656, −0.103);
ancient (0.6, −0.2)
Real shelf open acrylic rack; real shelf height above the table is CAN_STARTING_POSITION.md:190,208-210;
unmeasured (decides shelf6 vs shelf10; inert for pick scope, paper/AUDIT_INDEX.md:82-83; UPDATE_2026-08-
25.md:157-158
blocks place/nested claims)
Start pose fixed arm start (HARDCODED_START), real tool pose at start ≈ example.py:180; replay_harness.py:62-64;
cartesian_env.py:54
(0.367, 0.011, 0.090) base frame, ~constant across demos
Predicates see §3.5 (pick) and §6.2 (placed/contact/nested; replay_harness.py:52-56
NESTED_TOUCH_DIST = 0.066+0.015 = 0.081 m)
```

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2. Real-to-sim

2.1 Simulator pin

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item value pointer
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Engine Genesis, upstream github.com/Genesis-Embodied-AI/Genesis @ cluster/patches/GENESIS_PIN.md:1-16; commit
37b8d8f;
31951c3f (reports 0.2.1; ≈0.2.1 + ~270 upstream commits) + gripper_lab_2026-08-25.md:30-31
2-file headless-render patch; reproduce: clone, checkout
31951c3f, git apply
cluster/patches/genesis_0.2.1_headless_render.patch, pip
install -e .
Patch content genesis/ext/pyrender/renderer.py: per-env camera poses
cluster/patches/genesis_0.2.1_headless_render.patch
(env_poses) so one batched render serves N wrist cameras;
genesis/vis/camera.py: use_imshow=False gate on all
cv2.imshow/waitKey (headless)
Engine gate Genesis 1.2.1 rejected: 3.4× lower contact force, pinch CAN_STARTING_POSITION.md:391-418
grasp fails
Software python 3.10; torch 2.7.0+cu126; taichi 1.7.4; numpy 2.2.6; CLAUDE.md:7; cluster/install_lerobot.sh:7-
10,44-63;
cv2 4.11; one pip-only conda env serves genesis + lerobot + cluster_dv3_setup.sh:16;
SB3 (conda-forge libstdc++ poisoned the stack 07-30 → "NEVER cluster/install_r2dreamer.sh:8,30-31;
conda-install"); r2dreamer in a separate py3.11 venv (torch cluster/verify_env.sh:3-8,24-58
2.8.0+cu126)
Physics gs.init(precision="32", seed=0); SimOptions(dt=0.01, replay_harness.py:96-97,107-109;
substeps=8) → substep_dt 1.25 ms; 3 physics steps per env genesis_can_env.py:14,86,245-246;
step (30 Hz env clock); CPU backend for all training/eval trial_placements.json['world']
worlds (one world per process)
Contact solver defaults every geom carries MuJoCo-default sol_params (timeconst 0.02 gripper_lab_2026-08-
25.md:103-104,115-121
s, dampratio 1, dmin 0.90, dmax 0.95, width 1e-3, mid 0.5,
```

```
power 2); this tree only max()-clamps timeconst at
2·substep_dt = 0.0025 s (the old comment in
replay_harness.py:98-99 claiming 0.2.1 hard-wires it is
wrong for this tree) [CHECK 6]
Determinism replay is machine- and load-dependent (cross-box placed paper/METHODOLOGY.md:58-62; CLAUDE.md 07-20
BB
+4/75; heavy CPU load flips borderline demos) → same-machine
baselines, idle box, ×3
```

2.2 Scene (`can_pos_recovery/replay_harness.py::build_world`, lines 78-183)

```
object spec pointer
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Table (table=True) fixed box (0.419, 1.2, 0.05) at (0.3395, −0.1875, 0.025); :76,114-121
top TABLE_TOP_Z = 0.05; friction 0.5
Shelf ("box") box BOX_SIZE=(0.4,0.75,0.12) at BOX_POS=(0.75,−0.1875,0.05) :38-42,111-113; sim_variants.py:84-
88,111-122
→ BOX_TOP_Z = 0.11 (only 6 cm above the table in the base
world); ρ 1000 (~36 kg), friction 0.5; dynamic in base,
fixed=True under variants
Robot gen3_lite_2f_robotiq_85.urdf at (0,0,0.05), fixed; arm kp :81,122-124,171-179; sim_variants.py:27-30
[200,200,150,100,60,60], kv [20,20,15,10,6,6], force
±[50,50,50,20,20,20]; fingers kp 40 (finger_kp), kv 10,
force ±50 (finger_force)
Gripper 4 finger DOFs with mimic (tip = −0.676·bottom + 0.149) gen3_lite_2f_robotiq_85.urdf:313;
replay_harness.py:66-72;
enforced as equality constraints; reading 0..100 mapped gripper_lab_2026-08-25.md:109-110,127-134
linearly to driver θ ∈ [−0.09, 0.96] rad; collision = one
convex hull per finger mesh
Cans §1.4
Cameras (camera_rig=True) two 64×64 RGB cams: top at (0.40,−0.08,1.15) looking down replay_harness.py:141-170;
genesis_can_env.py:89,143-149
(up=(1,0,0), fov 68); wrist fov 80 attached to eef with
offset (0.10, 0, −0.03), pitched 30°, rolled 90°; obs image
= top ++ wrist = (64,64,6) u8. Video cam (not an obs)
640×560
```

2.3 Can starting-position and goal recovery

- Method: FK at the grasp instant (closure interval that lifts and releases near the goal; grasp = reach-bottom) → fingertip midpoint = can xy; validated against bag `tool_pose` to ~1 cm xy / ~1 mm z; `fk_recovered.json` covers all 96 trials (`CAN_STARTING_POSITION.md:40-53,181-187`; `can_pos_recovery/recover_can_position.py`, `fk_all_trials.py`).
- Placement search around the FK seed: `search_placements.py`, `batch_harness.py` (GPU B=32), `cpu_research.py` (CPU-parallel, ok-class by construction, ~25 s/rollout, 16-way) → `trial_placements.json` (`CAN_STARTING_POSITION.md:192-203`; `cpu_research.py:1-17`). Current file: 93 trials; 72 "solved" (`ok_batch 52, ok 20`); 20 `no_shelf_batch`, 1 `shelf_but_no_slide`; unsolved 255, 303 (`CLAUDE.md` 07-20 BB).
- Independent panel (2026-07-08) overturned a rigged negative control and a coverage overstatement (winner drift up to 17 cm; 60 % depended on goal relocation); fixes baked in: fail-labeled demos never receive relocated goals; nested requires picked (`CAN_STARTING_POSITION.md:254-305`; `CLAUDE.md`).
- World corrections found 2026-07-04: table added; can geometry/mass corrected (was 70×75 mm ρ 2000); finger gains restored; substeps 1→4→8 (pick/run 0.59→0.74, 16==8 converged) (`CAN_STARTING_POSITION.md:160-178,341-349`). Note doc says finger kp 100 "sweet spot" but the live world uses `finger_kp 40` (commit `e8a07d6`, 07-14) **[CHECK 7]**.
- `--ic-from-tape` (PREREG A3): the 12 IL demos without a recovered placement

(uids 233 259 262 266 267 275 278 301 319 321 329 333) reset to the can pose at frame 0 of their own stride-1 sim tape (`states[0][8:11]`, quat `[11:15]`), static goal; applied identically to all three sources so every source records from the same 66 ICs (`baselines/record_demos.py:662,731-757`; `PREREG:248-250`; `SESSION_LOG_2026-08-23_cluster.md:21`).
- Goal: §1.4. Whole-corpus replay at the final goal (75 success demos, base world): picked 61 / placed 47 / contact 15 / nested 16; fail-demo negative control nested 2/16 (`CLAUDE.md` 07-20; `METHODOLOGY.md:99-102`).

2.4 The human-command follower ("human" adapter)

The human demos are not re-encoded offline; they are **re-executed** inside the learners' MDP (§3) by a closed-loop waypoint follower, so every source is "recorded as executed" by one recorder.

```
item value pointer
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Source tape stride-1 sim tape from record_demos.py:416-421,656;
baselines/episodes_pick_phase_dppruned (the 66 IL demos, DP- real2sim_follower_lab_2026-08-23.md:82-88
pruned, §4.5); cmd = actions[:, :6] (real measured joints
used as targets), ref = states[1:, :6] (old-sim arm path),
grip = clip(actions[:,6])
Per decision target waypoint jt = min(j+3, n−1) (≤4 waypoints per record_demos.py:439-444
decision); a_arm = clip((cmd[jt] − target_now)/(4·0.025),
−1, 1); a_grip = grip[jt]·2 − 1
Arrival rule scan k from jt down to j; advance to k+1 if ‖q_meas − record_demos.py:427-437,448-470,668-671;
manifests
ref_k‖∞ < tol (meas) OR, with --arrival either, ‖target_env M/demos_v1/dH_either/manifest.json configs[0] (tol
0.025,
− cmd_k‖∞ < tol; tol = 0.025 rad; max_dwell = 8 decisions max_dwell 8, arrival either, dilation_cap 3.0, settle
25)
without arrival → forced jump (counted as a stall);
dilation_cap = 3.0 (step cap = ⌈3·n/4⌉ decisions →
adapter_exhausted); settle = 25 decisions holding the last
waypoint; dilation = decisions·4/n_src
Rule of record --arrival either (PREREG A6; lab: 49→55 local survivors, PREREG:272-276;
zero dwell stalls, identical path fidelity; commit 33d5bc9). real2sim_follower_lab_2026-08-23.md:23-31,132-
133,274-276;
The block's dH sets (matched_v2, matched_w3) are either; the SESSION_LOG:74-75
original 51-tape set (matched_v1) and the dDP teacher's
training data are meas (disclosed, A6)
Keep rule kept iff the hardened pick (§3.5) is re-earned; drops PREREG:241-243
reported as a source property (A1)
```

2.5 Sim variants (`baselines/sim_variants.py`)

Mechanics: `install(name)` monkeypatches `gs.Scene.add_entity` before build (robot gets `Rigid(gravity_compensation=gc)` and `pos.z += riser`; shelf gets `pos.z += shelf_dz`, `fixed=True`); `post_build(w, name)` sets arm kp/kv = BASE × mult, force range, optional `grasp_timeconst`, `goal_start_z += shelf_dz` (`:93-163`). Plumbed via `baselines/sim_variant_hook.py apply_pre/apply_post`; `'base'` = unpatched; stamped into every tape/manifest/sidecar/registry row; eval resolution "explicit CLI > checkpoint sidecar > 'base', mismatch = refuse" (`sim_variant_hook.py:1-10`; `record_demos.py:115-120,617-633,669`; `sbatch_dp.sh:195-196,300-303,475`; `sbatch_rlpd.sh:87,169-172,190,209`; `sbatch_r2dreamer.sh:209-212,252-258`; `wandb_eval.py:122-140`).

```
variant kp×/kv× gravity comp riser shelf_dz grasp_timeconst line
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-------------------------- ------------------------------------------------------------ --------------
base 1/1 0 0 0 — (engine default 0.02 s) :35
```

```
gc_kp4_riser3 4/2 1.0 0.03 0 — :51
gc_kp4_riser3_shelf6 (world of record for the corrected 4/2 1.0 0.03 0.06 (shelf top 0.17 world) — :67
block)
gc_kp4_riser3_shelf6_ts5 (pilot only) 4/2 1.0 0.03 0.06 0.005 s on the 4 finger geoms AND the picked can (Genesis
:69-78,138-156
averages a pair's sol_params, so both sides must move);
floor tmin = 2·substep_dt enforced by max()
gc_kp4_riser3_shelf10 (data-fit alternative) 4/2 1.0 0.03 0.10 — :80-82
others (gc, kp2, kp4, gc_kp8, *_urdf, riser2/4, fcap2, fmap) ablation ladder :36-60
```

kp×4 → arm kp `[800,800,600,400,240,240]`, kv×2 (`SESSION_LOG:136`). `post_build` prints but does **not assert** the 5-geom count (CRITIQUE Q8) **[CHECK 8]** (`sim_variants.py:156`; `CRITIQUE_decisions_2026-08-26.md:364-368`).

2.6 Physics defects found against real bag data, fixes, and fidelity

Source docs: `paper/real2sim_follower_lab_2026-08-23.md` (defects a–d) and `paper/gripper_lab_2026-08-25.md` (defect e). All numbers are local-devbox unless marked cluster.

```
defect evidence fix
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----------------------------------------------------------
(a) no arm gravity compensation + soft PD (sag τ_g/kp; ~4 cm real2sim…:166,171-176,460-462 gravity_comp 1.0, kp×4
/ kv×2
eef sag at the extended place pose, uid 242 sim eef 0.147 vs
real 0.191)
(b) bandwidth: leash-limited steady speed ≈0.75 rad/s vs :178-182 kp×4
real p99 0.6–1.35 rad/s
(c) robot mounted ≥3 cm too low relative to the table (real :184-195,208-215; sim_variants.py:46-49 riser 0.03
tool z reaches −0.019…+0.013 base-frame; sim finger hulls
bottom out at tool z ≈ +0.03) → "pressing" failures
295/299/254/256/286/298/330 and the 246 knock-over
(d) shelf modelled half-buried (top 6 cm above table; real :456-467; UPDATE_2026-08-25.md:91-96;
sim_variants.py:61-66 shelf_dz 0.06 (shelf10 = data-fit upper bound; real height
release tool z 0.17–0.21 ⇒ ≥12 cm) — masked in the base unmeasured)
world because the sag lowered the can onto the low shelf
(two cancelling errors: riser-only world nested 20→3)
(e) contact-solver timeconst 0.02 s = 8× the engine floor gripper_lab:25-35,46-53,103-121,184-216 grasp_timeconst
0.005 (ts5) — pilot only, not adopted for
2·substep_dt = 0.0025 s; penetration ∝ timeconst²; the old any result block (§2.7)
"finger reading map is miscalibrated" theory refuted (real
gripper over-travels into rubber/current limit at readings
0.667–1.007 on the same can)
```

Fidelity (replaying the real robot's recorded joint stream through the follower, 66 tapes, 39 332 frames):

```
metric base gc_kp4_riser3 / shelf6 +ts5 pointer
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------------------------------------------------------
joint tracking ‖e‖∞ p50 / p95 (rad) 0.047 / 0.100 (whole-corpus figure quoted as 0.045 / 0.090) 0.004 / 0.030
(0.003 / 0.029 in the 75-demo table; 0.0033 / 0.0035 / 0.0300 real2sim…:162-170,199-208,441-451; UPDATE:87-
89,104-108
0.0285 cluster)
frames beyond the leash (0.125 rad) 7.9 % (4.0 % in the 75-demo table) 0.1–0.2 % unchanged same
frames beyond the cap 87 % 10 % — real2sim…:199-208
tracking error at grip close 0.049 0.002 — same
pick re-earned under open-loop replay — 64/66 — real2sim…:41-46
full-task 75-demo replay 67/59/26/18/20/32 71/55/28/22/17/38 69/60/31/20/20/23 real2sim…:441-451;
gripper_lab:521-527,623-640
picked/placed/placed_v2/contact/nested/tipped
carry penetration med-of-max / p95 9.7–9.9 mm / 9.0 mm 9.9 mm 1.5 mm / 0.90 mm gripper_lab:623-640; UPDATE:104-
108
free tip-overs / 75 — 20 12 (10 recoveries vs 2 losses, McNemar p≈0.039) UPDATE:106; CRITIQUE_decisions:37-42
grip force med-of-max (obs[7]-related) 143 N 147 N 245 N (fingers stall at the surface holding full PD error;
gripper_lab:54-61,737-758
```

```
every reduction lever loses the grasp; engine has no
torsional friction/impratio)
human pick recreation (recorder, /66) 51 (meas) → 56 (either) old world 57 base-control vs 58 ts5 (cluster-
confirmed; ts5 rescues UPDATE:100-109; gripper_lab:459-486;
246, loses 316/254) — but 2 vs 1 discordant, McNemar p=1.0 CRITIQUE_decisions:37-42
random-teacher control 0/30 cluster, 0/90 local UPDATE:104; gripper_lab:603-607
E5 caveat ts5 pair-averaging also stiffens can↔goal-can and can↔shelf AUDIT_INDEX.md:75-77;
CRITIQUE_decisions:182-201
contacts 0.020→0.0125 s (1.6×) — the metric-bearing contacts
for contact/nested
```

[CHECK 9] the "57/66 vs 58/66 pick recreation" and "56/66 either" figures come from different runs (gripper-lab g_base control vs the 08-23 dH_either recording vs the 08-24 dH_w2 recording = 58/66 in `gc_kp4_riser3_shelf6`); state which one is meant.

2.7 Which world each block ran in

```
world sim_variant demo root seeds / cells pointer
--------------- ----------------------------------------------------------
-------------------------------------------------------
-----------------------------------------------------------
----------------------------------------------------------
old base baselines/matched_v2 (N=56) DP 10–14 (dH/dDP/dR2D); RLPD 10–13 (sparse+dense, + fails
analysis/results_table.py:10-20;
arms sparse); all r2dreamer cells (80–93, 99–108, 200–203, RESULTS_TABLE_2026-08-25.md:3; UPDATE:14;
300–303); N5/N15/N13/N16 CRITIQUE_decisions:389-391 (C3)
corrected gc_kp4_riser3_shelf6 (adopted by user 08-24 ~08:15, shelf6 baselines/matched_w3 (N=58; dH + dDP only,
no corrected- DP 20–29; RLPD 20–29 (sparse+dense); N7 density 20–22; N14 SESSION_LOG:129,141,158-160;
M/matched_w3/MATCHED_SETS.json
pending the real shelf measurement) world dR2D teacher) split halves 40–42; r2d pilot 60/61 (0/2 at TL 1200)
corrected + ts5 gc_kp4_riser3_shelf6_ts5 (commit 7c8195d "world of record matched_w4_pilot/dH only (dH_w4
recording 08-26 07:50) DP pilot dp_pilotw4 only; no result block; matched_w4 proper SESSION_LOG:153-156;
results_table.py:19-20;
from 08-26") never built; no kept count/readout for dH_w4 recorded in any POSTMAINT_COMMANDS.md:118-126
note [CHECK 10]
```

---

3. The MDP and the contract-v1 recorder (`baselines/record_demos.py`)

3.1 Environment

`FullTaskEnv(backend='cpu', max_steps=1200, scope='pick', action_mode='delta_joint', delta_cap=0.025, delta_leash_mult=5.0, action_repeat=4, delta_ref='target', camera_rig=True, pick_shaping=False)` (`record_demos.py:94-103,219-231`; `baselines/rl/full_env.py:265-270,310`). Asserted at construction: scope/mode/ref, repeat and cap, leash == 0.125, no shaping/hold reward, inner env never truncates (`genv.max_steps = 1e9`, the #26 bug guard) (`record_demos.py:226-230`; `full_env.py:351`). `sim_variant` applied pre/post build and stamped (`:221,231,781`).

3.2 Observation (17-dim state + images)

`state = [q[0:6], gripper motor 0..1, grip effort, can xyz, can quat wxyz, goal xy]` (`genesis_can_env.py:298-313`; `pick_env.py:36`). Gripper motor = `1 − (θ − (−0.09))/(0.96 − (−0.09))`; grip effort = |control force| on the two bottom finger drivers ("sim analog of motor current"; its scale moves with the world variant: carry median 18.4 base → 26–28 ts5, `gripper_lab:772-776`). lerobot split: `observation.state = states[:8]` (proprio), `observation.environment_state = states[8:]` (can pose + goal) (`convert_to_lerobot.py:81-89`). Images `(64,64,6)` u8 top++wrist (§2.2); WM learners use images only (`cnn_keys: image`, `mlp_keys: '$^'`), DP and RLPD use state only. **[CHECK 11]** `genesis_can_env.py:10` docstring still says `float32[16]`.

3.3 Action space and integrator

`Box(−1,1,(7,))` (`full_env.py:392`). Per sim step (`full_env.py:560-578`):

```
d  = clip(a[:6], -1, 1) * delta_cap              # 0.025 rad / sim step

sp = clip(target + d, ARM_LO, ARM_HI)            # delta_ref='target': integrate onto the running target

target = q_meas + clip(sp - q_meas, -leash, +leash)   # leash 0.125 rad around the measured joints

grip_phys = (clip(a[6], -1, 1) + 1) / 2          # absolute 0..1
```

`action_repeat=4`: the same `a` is applied on 4 consecutive sim steps (advance ≤ 4·a·cap), rewards summed, loop breaks on terminated/truncated (`full_env.py:531-542`). Target re-seeds from measured q on every reset (`:402-412`). Calibration: cap 0.025 = the demos' p99 per-frame commanded delta (44°/s at saturation); leash 5×cap = 0.125 ≈ the demos' PD-lead p99 (0.126) (`full_env.py:300-302`; `paper/METHODOLOGY.md:233-244`). Decision clock = 4 sim steps = 7.5 Hz. DP emits absolute window-end targets and is executed through the same integrator at eval ("hold-4", §6.2).

3.4 Horizon and termination

- `max_steps = 1200` sim steps = 300 decisions; `truncated = (not terminated) and t ≥ max_steps` (`full_env.py:689`); the tape ends exactly at the terminal or the cap (`record_demos.py:156,318-325`).
- **Tip rule**: after each sim step, if `grip_phys < GRIP_OPEN (0.3)` and can tilt > `TIP_DEG (60°)` → `terminated`, `info['tipped']`, reward += `TIP_PENALTY`. **`TIP_PENALTY = 0.0` for pick/full scope** (`PLACE_TIP_PENALTY = −0.25` place scope only). PREREG §2 says r=0. The "−0.5" in the task brief/CLAUDE.md history is the July Cartesian env; not current **[CHECK 12]** (`full_env.py:214-218,673-688,705`). Census basis: past 60° the can never recovers (31/32); grip-open guard needed because demos carry the can pitched >60° in hand (`CLAUDE.md`; `METHODOLOGY.md:120-124`).
- The **eval** harness uses `GenesisCanEnv` (no tip termination, runs the full horizon) — §6.2.

3.5 Pick predicate ("hardened lift")

All of `can_z > pick_z` AND `grip·100 > GP_CLOSE (30)` AND `‖eef − can‖ < PICK_EEF_DIST`

`(0.20 m)` sustained `PICK_SUSTAIN = 10` consecutive sim steps; `eef` = wrist link position; `pick_z = table_top + can_h/2 + 0.05 = 0.05 + 0.0505 + 0.05 = 0.1505` (`genesis_can_env.py:48-65,255-265`; `replay_harness.py:184`; census header "pick_z 0.1505"). Pick scope terminates on the sim step the grant lands (`full_env.py:627-633`); no settle. Hardening history and the frozen-policy fairness certification: `paper/METHODOLOGY.md:109-118`.

3.6 Rewards

- Sparse: `STAGE_REWARD = {picked 1.0, placed 1.0, contact 2.0, nested 4.0}`, paid once per first grant; pick scope → +1 and terminate (`full_env.py:57,593-606,627-633`). Tapes carry sparse env rewards only ("shaping is NOT baked in", `record_demos.py:22`).
- **Fast-pick 2.0 quirk**: if the pick grant and the placed grant land in the same repeat-4 window (`GenesisCanEnv.step` sets `_picked` then `_placed` in the same sim step when the can is already in the shelf footprint/z-band), the terminal row carries reward 2.0. Seen on r2d-teacher tapes (~68-step picks); `to_dreamer_native.py` accepts `{0,1,2}` and refuses positive reward before the terminal row; matched_v2 `r2d/dR2D` total_reward 64 for 56 tapes (8 tapes pay +2) (`baselines/rl/to_dreamer_native.py:65-72`; `genesis_can_env.py:265-267`; `SESSION_LOG:114`). Mechanism reading is from code, not from a comment **[CHECK 13]**.
- Dense (potential-based): `r' = r + γ·φ(s') − φ(s)`, `φ = −PICK_SHAPING_SCALE·‖eef − can‖` with `SCALE = 2.0`, **φ(terminal) = 0** (`pick_shaping_terminal_zero`), γ = the learner's discount (RLPD 0.998 via `pick_shaping_gamma=args.gamma`, asserted equal; r2d 1 − 1/horizon = 0.997 after the patch; dv3 0.997); demo transitions relabelled offline from the recorded `eef_pos` with the same potential (`full_env.py:95-102,261-262,333,543-557`; `train_rlpd.py:99,245,255,308-312`; `cluster/patches/r2dreamer_final_rr.patch` hunks 1–2; PREREG §2).

3.7 `terminal_from_tape` and the terminal guard

`full_env.terminal_from_tape(tape, …) → {t_term, kind ∈ pick|tip|other, reward, layout}`: contract-v1 tapes → reads `terminated/picked/tipped` (`full_env.py:152-168`); legacy stride-1 tapes → recomputes the pick proxy + tip rule, warns that the hardened eef distance is unavailable (`:173-211`). Every legacy encoder/converter has `--demo-terminal-guard` default **on** (tip → done, r=0, later frames dropped; pick → done; cap → bootstrap) (`train_rlpd.py:145-153`; `train_sacfd_full.py:622`; `to_dreamer_demos.py:70`; PREREG §4.3). This guard is the fix for the 08-19 `dDP_RLPD` 0/6 collapse (fail tapes without terminal flags → critic bootstrapped through states the env would have terminated; N1).

3.8 Tape contract v1 — keys and manifest

One npz per episode, filename = rollout index ≥ 100000 (`rollout_base + 1000·shard_idx`, never a human uid; `record_demos.py:774-777,804`). Keys (`:16-33,329-341`): `states (n,17)`, `final_state (17,)`, `actions_delta (n,7)` (the executed normalized decision — the learners' action space), `actions (n,7)` (absolute window-end command: 6 joint targets rad + grip 0..1), `rewards, terminated, truncated, picked, placed, contact, nested, tipped (n,)`, `eef_pos (n+1,3)` (tool tip), `images (n+1,64,64,6)`, `sim_states (m,17)`, `sim_actions (m,7)` (per-sim-step sub-tape, n ≤ m ≤ 4n), `end_reason`, `n`. Scalars (`:115-120`): `uid, ic_uid, label ('success' iff picked on the last row), stage, teacher, teacher_ckpt, act_mode, sim_variant, action_repeat=4, delta_cap, delta_leash, delta_ref='target', pick_z, n, git_sha, env_class='FullTaskEnv', recorder='record_demos.py v1', contract='v1', end_reason, scope, max_sim_steps, verify`. Validator: exactly one of terminated/truncated on the last row and never earlier (`:125-200`). Manifest (`:830-846`, merged `:599-635`): full argparse config incl. `tol, max_dwell, arrival, dilation_cap, settle, attempts, verify, seed, shard, git_sha, teacher_ckpt`; `n_rollouts, n_kept, n_fail, rejected_by_verify, teacher_success_rate_first_attempt, yield_frac`; per-rollout `records` (ic_uid, ic_can_xy, outcome, decisions, sim_steps, seconds, kept, end_reason, attempt, verify, dilation…); `content_sha256` (sha over sorted `name bytes + file bytes`), `built`. Fails go to `<outdir>_fails/` (same schema, `label='fail'`); the 5xxxxx stems are assigned by the set builder (`make_matched_sets.py:151-161`: `500000 + stem % 100000`), 6xxxxx by `make_succ_control_set.py`.

3.9 Adapters and guards

```
adapter behaviour pointer
-------------- ------------------------------------------------------------- -----------------------------------
human §2.4; attempts forced to 1; verify off (the keep predicate record_demos.py:395-470,694-695,742
is the pick itself)
dp lerobot DP queried once per decision on the current obs; :473-504,654,663; manifests
absolute target q* → a_arm = clip((q* −
target_now)/(4·cap)); grip = chunk value ·2−1; --mode sample
(default) / mode (reseeds per episode); policy on GPU if
visible, sim CPU (A4); --attempts 3 --verify
r2d champion loader, native delta actions, asserts cfg :507-582
action_mode/repeat/cap/leash/size; --mode sample --attempts
3 --verify; --torch-threads 2
random uniform [−1,1]^7 per decision (rng seed+7919); warns if kept :388-392,849-850
> n/30
--verify open-loop replay of actions_delta from a fresh reset of the :362-371,685-687,798-803
same IC must re-earn the pick; failures routed to fails with
outcome='verify_failed'; pick scope only
--ic-from-tape §2.3 :662,731-757
--sim-variant default base; merge refuses mixed variants :669,619-622
```

---

4. Demonstration sources and sets

4.1 Recordings (all by `record_demos.py v1`, repeat 4, cap 0.025, leash 0.125, 1200 sim steps, `src = baselines/episodes_pick_phase_dppruned`, `--ic-from-tape`, 66 ICs cycled)

```
recording world teacher / adapter rollouts → kept first-attempt rate verify rejects content sha (prefix) pointer
-------------------- -------------------- -------------------------------------------------------------
------------------------------------------------------------ ------------------
------------------------------------- -------------------
-----------------------------------------------------------
dH (meas) base human, --arrival meas, git 91cd7b1/bde726d 66 → 51 — — 58fd89bc
paper/final_rr_sets_2026-08-23/recorder_manifest_dH.json
dH_either base human, --arrival either, git 33d5bc9 66 → 56 0.848 — 4d4ea08f
…/v2/recorder_manifest_dH_either.json;
M/demos_v1/dH_either/manifest.json
dDP base dp, ckpt baselines/outputs/dp_pilot/dH_DP_s0/checkpoints/020 80 → 64 0.818 2 (uids 276, 317 →
dDP_fails_verifyrej) c33a4e2e M/demos_v1/dDP/manifest.json; SESSION_LOG:58,61; PREREG A6
000/pretrained_model (DP-r4 pilot s0, sel 13/15; PREREG §3.1 (teacher trained on the 51 meas tapes — a
superset/variant
"median in-dist seed" → with 2 pilot seeds the lower), relation to the block's dH, disclosed)
--mode sample --attempts 3 --verify, git 1dcd268
```

```
dDP_fails base same teacher 16 → 14 files: 8 env_truncated at 300 decisions (uids 258 0.0 2 41a572f3
M/demos_v1/dDP_fails/manifest.json
266 267 298 299 304 326 327) + 6 lying-can tapes of 1
decision (234×3, 318×3); 2 verify-rejected
dR2Dprov base r2d, ckpt C/r2dreamer/runs/pick_v5d4c_delta_shaped_dH_s51/BE 73 → 64 (8 tipped, 1 verify-failed uid
333) not stamped 1 16f62a43 …/recorder_manifest_dR2Dprov.json;
ST_selected.pt (PREREG A2 fallback, not AUDIT_prelaunch_2026-08-23.md:207
CHAMPION_1576820.pt), --mode sample --attempts 3 --verify,
git 3167d12/f6d937f
_smoke_negctl base random, n=3 3 → 0 (2 truncated, 1 tipped) 0.0 — — M/demos_v1/_smoke_negctl/manifest.json;
PREREG A5 (n=3, not
the registered 30)
dH_w2 gc_kp4_riser3_shelf6 human, either, git e306ec9 66 → 58 (drops: tipped 234, 318; env_truncated 245 246 286
0.879 — 07ce3f36 M/matched_w3/dH/manifest.json; SESSION_LOG:129
293 295 300); dilation p50 0.99, max 1.07
dDP_w2 gc_kp4_riser3_shelf6 dp, ckpt baselines/outputs/dp_pilotw2/dH_DP_s0/checkpoints/1 76 → 64 (7 tipped, 5
truncated) 0.879 0 1da70d3f M/matched_w3/dDP/manifest.json; SESSION_LOG:141
00000/pretrained_model (pilot hold 14/15, rnd 18/30),
attempts 3 --verify, git 2061f08
corrected-world dR2D — does not exist (no corrected-world r2d teacher; the M/matched_w3/MATCHED_SETS.json
NOTE_2026-08-25;
matched_w3/dR2D row was a byte-identical copy of dDP and was CRITIQUE_launch_plan_2026-08-25.md:210-234
purged, BL-7)
dHunpruned{,_either} base human on episodes_pick_phase_all (control for §4.5 pruning) recorded; no result appears
in any results file [CHECK 14] SESSION_LOG:60,94
```

dH drop reasons vs PREREG A1 wording ("2 lying-can, 1 knock-over, 5 over-horizon, 7 over-cap" for the 15 meas drops): recorder outcomes are 3 tipped (234, 318 lying-can ICs; 246 knock-over), 10 `env_truncated` (1200-step horizon), 2 `adapter_exhausted` (dilation cap 3.0: 257, 298) — the 5/7 split does not match the 10/2 outcome codes **[CHECK 15]** (`recorder_manifest_dH.json records[]`; `PREREG:241-243`). Lying-can ICs 234 and 318 are the two human demos whose recorded can starts on its side; they are unachievable in the MDP and also make uid 234 the `sel` ceiling (§6.1).

Teacher circularity (disclosure): both model teachers descend from dH (dDP = DP trained on the 51-tape meas dH; dR2D = r2dreamer dense trained on the 08-19 human set) (PREREG §3.1, §10).

4.2 Matched sets (`baselines/make_matched_sets.py`)

Rule: `N = min(success counts, --cap-n 66)`; stratified round-robin over `ic_uid` with common capacity `c(u) = min over sources` so all sets cover the same uids with the same multiplicity (`ic_multiset_identical: true`); `rng = default_rng(seed=0)`; tapes hard-linked; refuses non-v1 tapes, label/picked mismatch, unterminated tapes, mixed `sim_variant` (`:64-78,103-142,167,183,207,211`). Frames are not matched (intrinsic; reported). `content_sha256` over sorted `name + bytes` (`:94-100`); every sbatch asserts the sha of the set it consumes. Fails arms: `F ≤ ⌊0.30·N/0.70⌋`; lying-can fail tapes (`n ≤ 1 and tipped`) excluded since commit ca05472 (W5) — hence matched_v1 fails arms carry 14 fails and matched_v2 carry 8 (`:222-231`). Per-learner derivations are built separately: lerobot via `convert_to_lerobot.py <set> <set>/lerobot 8 4 none image` (fps = 30/4 = 7.5, gated against `meta/info.json`), dreamer-native via `to_dreamer_native.py` (§4.7); `sbatch_dp.sh:14,221` wrongly says make_matched_sets builds them **[CHECK 16]**.

```
root built / git N dH sha dDP sha dR2D sha dDPfails (N) dR2DDPfails (N) world pointer
-------------------------------------- ------------------------------------- -- -------- -------- --------
-------------------- --------------- -------------------- ------------------------------------------------
matched_v1 08-23 13:53 / 0e1fda19 51 58fd89bc 9be55219 fe93dbdd c84fd1c1 (65) 4bf50466 (65) base
paper/final_rr_sets_2026-08-23/MATCHED_SETS.json
```

```
matched_v2 (block of record, old world) 08-23 14:17 / ca054726 56 4d4ea08f 299825ad 10cbc9b2 89609cc1 (64 = 56+8)
b222df06 (64) base …/v2/MATCHED_SETS.json; M/matched_v2/
matched_w3 (corrected world) 08-24 11:46 / 2061f082 58 07ce3f36 d3bf95f6 — none none gc_kp4_riser3_shelf6
M/matched_w3/MATCHED_SETS.json
matched_w4 never built (only matched_w4_pilot/dH) — ts5 SESSION_LOG:156
```

Provenance chain for the WM fails arm (state once): RLPD manifest sha `b222df06…` = `repeat.json.src_sha` → native sha `3a6df556…` → registry fingerprint `c5ae239b…` (`AUDIT_results_2026-08-28.md §5`).

4.3 Census (`analysis/characterize_demo_sets.py`; constants pick_z 0.1505, cap 0.025, leash 0.125, tip 60°/grip<0.3)

```
metric v1 dH / dDP / dR2D v2 dH / dDP / dR2D v2 dDPfails / dR2DDPfails w3 dH / dDP pointer
----------------------------------------------------------- ------------------------- -------------------------
------------------------------------------------------ ----------------
----------------------------------------
tapes 51 56 64 / 64 (8 fails each) 58 census.md per root
transitions (decisions) 6491 / 6294 / 861 7002 / 6909 / 954 9309 / 3354 6927 / 7285
tape length p50 (decisions) 113 / 113 / 17 115 / 114 / 17 122 / 17 (fail len 300) 110 / 120
tape length mean / max 127,285 / 123,233 / 17,24 125,284 / 123,233 / 17,24 145,300 / 52,300 119,246 / 126,235
fail share of buffer (rows) — 0 0.258 / 0.716 (=25.8 % / 71.6 %; PAPER_NOTES' 74/28 was 0 AUDIT_results:96;
M/matched_v2/census.md
wrong)
unique ICs / grid cells (of 16) 50 / 13 55 / 13 56 / 13 57 / 13
reward density .0079 / .0081 / .0592 .0080 / .0081 / .0587 .0060 / .0167 .0084 / .0080
decisions at the cap .055 / .010 / .048 .069 / .010 / .047 .054 / .024
human time dilation p50 / max (blank) 1.03 / 1.65 0.99 / 1.07
tapes tipped at t0 0 (fails arms 6) 0 0 0
all tapes: contract v1, repeat stamp 4, terminal on last row yes yes yes yes
only, images present
```

The "decisions per tape" summary used in prose (dR2D ~17, dDP ~123, dH ~115) is the v2 p50/mean mix. Fail tapes are 300–301 rows each (cap-truncated, bootstrapped), 12× the median dR2D success tape and 3× the WM training `time_limit`; they are 70 % of the fails arm's demo rows (`AUDIT_results §1`).

4.4 Split halves (`baselines/make_split_halves.py`, N14)

Group tapes by `ic_uid`, shuffle uid list with `default_rng(seed=0)`, alternate whole ICs A/B; per-half manifest with `split_half, split_seed, sha` (`:28-56`). Run on matched_w3 dH and dDP: "58 tapes over 58 ICs → A 29 / B 29" (`SESSION_LOG:157`). Halves' shas not captured locally **[CHECK 17]**.

4.5 Pruning

- Leading-idle pruning applied ONCE to the human stride-1 tapes before re-recording (`make_dp_pruned.py`: collapse idle runs with action delta < 1e-3 before `j_pick − 150` frames; 29.6 % of frames) — this is `baselines/episodes_pick_phase_dppruned`, the `src` of every recording; the unpruned control is `dHunpruned` (`make_dp_pruned.py:7-15,75-81`; `PAPER_PLAN.md:400-407`; PREREG §3.1).
- Post-hoc density control sets (N7, `baselines/make_pruned_bc_set.py:46-49`): keep decision iff `max|a_arm| ≥ eps` OR `|Δa_grip| > 5e-3`; terminal always kept; BC-only (sim sub-tape dropped). Measured at eps 1e-3 on matched_w3/dH: 6927 → 6116 decisions (**11.7 % removed**; 414 of the removed had grip closed). The 1e-2 fraction ("22 %") exists in no artifact — source or delete

[CHECK 18] (`SESSION_LOG:159`; `AUDIT_results:84,114`).

4.6 Fails arms and controls

- `dDPfails` = 56 dDP successes + 8 dDP fails (episode share 0.125; row share 25.8 %); `dR2DDPfails` = 56 dR2D successes + the same 8 fails (row share 71.6 %). The 8 fail tapes are byte-identical across the RLPD and WM arms (`AUDIT_results §0`).
- N16 controls (`make_succ_control_set.py`, PREREG A10): `dR2DDPsucc` = dR2D + the 8 longest dDP success tapes (6xxxxx stems; 1521 added rows vs the fails' 2400 → `length_match_fraction 0.634`; N 64) (`M/matched_v2/dR2DDPsucc/manifest.json`); `dR2Ddup13` = dR2D at `demo_duplicate 13` (ring share ≈27 % = fails arm) via `sbatch_r2dreamer.sh:199,236`. Predictions registered in `PAPER_NOTES.md` N16 (`:419-441`).
- Exposure arithmetic of record (supersedes every "1–3 %" / "1/17th" figure): with `demo_reinject_every 150000 × demo_duplicate 4` over a 3e6-step budget → 19 re-injections; steady-state ring share dR2D ≈ 9.8 %, fails arm ≈ 26.7 % (fail rows 18.8 %); RLPD 50 % demo per batch, fail rows 35.8 % → exposure ratio ≈ 2× (`AUDIT_results §1`).

UPDATE 08-29. Same-source fails arms only (user rule 08-28: never mix sources): `dHHfails` = 56 dH successes + 8 dH fail tapes; `dDPfails` = 56 dDP successes + 8 dDP fail tapes (`baselines/make_samesource_fails_arm.py`; fail share 0.30 by tape was requested but only 8 fail tapes per source survive the lying-can/success-label filters → share 0.125 by tape, fail rows 2145 / ~2300 = 23–25 % of rows; 7 truncated + 1 tip-terminated (dH), 8 truncated (dDP)). Row-matched same-source controls `dHsucc_dup` / `dDPsucc_dup` = the 56 successes + the 8 LONGEST own successes duplicated (6xxxxx stems; `baselines/make_succ_control_set.py --donor <same arm>`; added rows 1755 / 1521). Reading rule (A17): a fails effect is claimed only if +fails differs from +succ_dup in the same direction on both sources. Census per source in `paper/tape_census_2026-08-29.txt` (`baselines/diagnostics/tape_census.py`): dH vs dDP identical on N, rows, terminal rows, saturation (≤6 % on any joint), state ranges; dDP tapes carry `verify=pass` (open-loop replay at record time), dH `verify=n/a`. The mixed-source `dR2DDPfails` / `dR2DDPsucc` arms are mechanism cells only (N15, A12).

4.7 Dreamer-native conversion (`baselines/rl/to_dreamer_native.py`)

`action[t] = actions_delta[t−1]` (zeros at 0); `reward[t] = rewards[t−1] × terminal_reward` (zeros at 0); `is_terminal[-1] = terminated[-1]` (cap-truncated → False → bootstraps); `discount = 1 − is_terminal`; `is_first/is_last`; images T = n+1; refuses shaping, rewards ∉ {0,1,2}, positive reward before the terminal, repeat-stamp mismatch, mixed cap; writes `repeat.json` (action_repeat, contract, delta_cap, scope, terminal_reward, src_sha, src_manifest_sha, census) (`:49-83,111-131,146-156`). r2dreamer dirs built with `--terminal-reward 1` (prefill multiplies by reward_scale 100, asserted); dv3 dirs with `--terminal-reward 100` — same effect, different file (`SESSION_LOG:100,111`; `r2dreamer_final_rr.patch` hunk 3).

5. Learners

5.1 DP — lerobot Diffusion Policy (state-only)

```
knob value pointer
-------------------------------------------------------
-------------------------------------------------------------
------------------------------------------------------------- --------------------
Code lerobot fork dirkmcpherson/lerobot branch genesis-fixes cluster/install_lerobot.sh:17-26,49,60-68
(lerobot 0.4.5 + image_writer mkdir fix; cluster checkout
b63920e), torchcodec 0.3.*
Inputs observation.state (8) + observation.environment_state (9); sbatch_dp.sh:82-85,420;
convert_to_lerobot.py:81-90
no images (cameras none)
Action 7 = 6 absolute joint targets (window end) + grip 0..1; fps convert_to_lerobot.py:42-51,90
7.5
Train lerobot-train --policy.type=diffusion --seed=$SEED sbatch_dp.sh:187,198,203-204,448-454
--batch_size=64 --steps=100000 --save_freq=20000
DiffusionConfig (lerobot defaults, nothing overridden) n_obs_steps 2; horizon 16; n_action_steps 8; cluster fork
src/lerobot/policies/diffusion/configuration_di
drop_n_last_frames 7; normalization MIN_MAX (state, action); ffusion.py:104-163; modeling_diffusion.py:204-207
UNet down_dims (512,1024,2048), kernel 5, n_groups 8,
diffusion_step_embed 128, FiLM; DDPM 100 train timesteps,
squaredcos_cap_v2, β 1e-4→0.02, ε-prediction, clip_sample
1.0; num_inference_steps = 100; Adam lr 1e-4, betas (0.95,
0.999), wd 1e-6; cosine schedule, warmup 500; vision
backbone not instantiated
Checkpoints K=5 at 20k…100k, each with dp_sidecar.json (action_repeat 4, sbatch_dp.sh:459-482
demo_sha, sim_variant, git, node, config)
Execution hold-4 through the env integrator (§3.3); chunk of 8 wandb_eval.py:105-109,271-326; dp_runner.py:39-50;
decisions between replans; sampled actions, diffusion noise eval_sweep.sh:118-128
seeded per episode (--seed k); policy on GPU if visible
Selection cluster/dp_select_confirm.sh: score 5 ckpts on sel, best dp_select_confirm.sh:23-50; PREREG A7
(ties → later, -ge) confirmed on hold+rnd, final also on
hold+rnd; DP-HEADLINE line + sweep/HEADLINE.txt
Seeds old world (matched_v2): dH/dDP/dR2D × 10–14 (n=5); corrected M/RUN_REGISTRY.jsonl; results_table.py:10-20
(matched_w3): dH/dDP × 20–29 (n=10); N7 density arms 20–22;
N14 halves 40–42; pilots s0/s1
Not run the PREREG §1 300k "epoch-matched view" (no registry row, no PREREG:32
sbatch path) [CHECK 19]
Resources `-p gpu,preempt --requeue --gres=gpu:1 --constraint="l40s\ a100" -n 8 --mem 48g --time 14h`
sbatch_dp.sh:154-167
```

5.2 RLPD — SB3 SAC subclass (`baselines/rl/train_rlpd.py`, `rlpd_sac.py`, `cluster/sbatch_rlpd.sh`)

```
knob value pointer
--------------------------------- -------------------------------------------------------------
------------------------------------------------------- ---- -------------------------------- --------------------
Base stable-baselines3 2.8.0 (asserted) rlpd_sac.py:38-40; sbatch_rlpd.sh:240
Critic ensemble E = 10 LayerNorm critics (vectorised EnsembleLinear), LN train_rlpd.py:58; rlpd_sac.py:91-126,372
after each hidden Linear, shared affine (--per-member-ln
off)
Target subset Z = 2 random target critics, min → Bellman target, redrawn train_rlpd.py:60; rlpd_sac.py:258-260
every grad step
Networks π (256,256), Q (256,256) rlpd_sac.py:345,371
UTD 10 critic grad steps per decision; actor + α once per train_rlpd.py:55; rlpd_sac.py:240-241,282-301,363-364
decision
Batch 256 = 128 online + 128 demo (symmetric sampling from an train_rlpd.py:61; rlpd_sac.py:232,243-249,360
immutable demo buffer)
γ 0.998 train_rlpd.py:99
Entropy ent_coef auto, target −dim/2 = −3.5, entropy backup off train_rlpd.py:63,103,107; rlpd_sac.py:269-
271,350-352
lr / τ / buffer / learning_starts 3e-4 / 0.005 / 300 000 online / 1 000 rlpd_sac.py:357-361
Budget 100 000 decisions = 400 000 sim steps sbatch_rlpd.sh:33,80-83,89
(BUDGET_UNIT=decisions); train horizon 1200 sim steps; eval
horizon 1200
Env FullTaskEnv(scope='pick', action_mode='delta_joint', train_rlpd.py:238-246
action_repeat=4, delta_ref='target', pick_shaping=…,
pick_shaping_gamma=0.998, pick_shaping_terminal_zero=True),
cap 0.025, leash 0.125
Demo format --demo-format native: (states[t], actions_delta[t], train_rlpd.py:145-160,307-312; sbatch_rlpd.sh:191
rewards[t] (+shaping from eef_pos when dense), next =
states[t+1] or final_state, done = terminated[t]);
```

```
truncation bootstraps;
sim_variant/repeat/cap/leash/delta_ref stamps checked;
--demo-terminal-guard on for legacy tapes only
Dense --pick-shaping on --demo-shaping on --pick-shaping-terminal- sbatch_rlpd.sh:186-190
zero on (§3.6)
Demo RNG torch.Generator().manual_seed(seed) (post-2fbed2a fix) rlpd_sac.py:179-182
Checkpoints K=5 via ArchiveCheckpointCallback at 0.2…1.0 of steps → train_rlpd.py:171-175,440-471,479-480
ckpt_020…ckpt_100/rlpd_ckpt.zip + .action_mode.json sidecar;
rlpd_final.zip
Eval action deterministic (predict(deterministic=True)), policy on CPU wandb_eval.py:236,263; eval_sweep.sh:126
Selection same as DP (5 × sel → best, ties → later → hold+rnd; SWEEP- sbatch_rlpd.sh:248-306
HEADLINE)
Seeds old world (matched_v2): dH/dDP/dR2D × sparse+dense × 10–13 registry
(n=4); dDPfails, dR2DDPfails sparse 10–13; corrected
(matched_w3): dH/dDP × sparse+dense × 20–29 (n=10). (Task
brief's "10–14" applies to DP, not RLPD [CHECK 20])
Resources `-p gpu,preempt --requeue --gres=gpu:1 --constraint="l40s\ a100\ l40\ h200" -n 8 --mem 48g --time 12h`
sbatch_rlpd.sh:62-72
Positive control none passing (PREREG §10 disclosure); recipe tuned under PREREG §10; PAPER_NOTES.md N6
old-world dynamics (N6/N10)
```

UPDATE 08-29 — recipe restart (PREREG A17). All γ = 0.998 runs (seeds 10–19, 24–29; waves `final`, `w2final`) are superseded, diagnostic-only (E9): 69/74 diverged (critic loss 10^2–10^5, Q-watchdog trips, hold ≤ 3/15). Fix factorial (seeds 30–32, old world, sparse): γ 0.99/UTD 10 (**g99**) — dH 3/3 stable (max critic loss 0.016–0.027, 0 trips, hold 14/14/14, rnd 20/21/19); dDP hold 10/13/14, rnd 14/12/20 with 2/3 seeds diverging (max critic loss 22, 84). γ 0.99/UTD 5 same picture; γ 0.998/UTD 5 diverges → γ is the cause. **Matrix recipe = γ 0.99, UTD 10, E 10, Z 2, 50/50 demo batches, LayerNorm critics, `ent_coef auto`, target entropy −3.5** (Ball et al. 2023 values except UTD 10 vs 20). Seeds (fresh ids, A9): dH/dDP sparse 30–37; dHHfails/dDPfails sparse 30–37; dH/dDP dense 30–35; dHsucc_dup/dDPsucc_dup sparse 30–33. Launcher knobs `GAMMA`, `UTD` (`cluster/sbatch_rlpd.sh`, registry keys). Dense arms relabel demo rows with the run's γ at load (`train_rlpd.py:311`) — γ 0.998 and 0.99 dense runs never share a table. Q-watchdog is warn-only (`rlpd_sac.py:318-324`) and its 2.0 threshold is invalid on dense arms (potential shaping lifts Q); health on dense arms = max critic loss + final ≈ selected. Statistic (A16, pre-registered before any s33–37 readout): divergence rate (Fisher) + LAST-checkpoint rnd-30 success (Welch + exact permutation, `analysis/stats.py`) — result 08-29 20:00: LAST rnd dH 0.700 vs dDP 0.494 (+0.206, CI [+0.105, +0.306], p 0.001), divergence 1/8 vs 6/8 (Fisher 0.041), see RESULTS_for_writing §2; hold is reported, not the statistic (dH at ceiling 14/15; hold ⊂ training ICs, A8). Time limit 18 h (one 12 h job lost its final-ckpt eval on a slow node). Env fix timing: jobs started before 08-28 13:25 carried the +2 double pick grant (`full_env.py`, `n_double_grant` normalised in the native loaders) — g99 runs are all post-fix.

5.3 r2dreamer (world model, pixels)

What it is: R2-Dreamer (Morihira et al., ICLR 2026) — a decoder-free, augmentation-free DreamerV3-family world model with a Barlow-Twins-style redundancy-reduction representation loss, shipped with its own PyTorch DreamerV3 reproduction (`model.rep_loss ∈ {r2dreamer, dreamer, infonce, dreamerpro}`); vs DreamerV3: no decoder, BlockGRU (8 blocks), barlow loss 0.05 (`lambd 5e-4`), and a **second, replay-based critic loss** (`repval`, weight 0.3) that backs up a λ-return along the recorded trajectory (demo/fail tapes included) bootstrapped with the imagined return (`C/r2dreamer/README.md`;

`_base_.yaml:23,36,45,51,76`; `dreamer.py:539-563`, disc `:549`, clamp `:552-553`; `AUDIT_results §1`). Port: cluster tree at commit `c25eb3b` + working-tree edits captured in `cluster/patches/r2dreamer_final_rr.patch` and `cluster/r2dreamer_port.tar.gz` (`AUDIT_sources_2026-08-23.md:248-287`).

Patch (`cluster/patches/r2dreamer_final_rr.patch`): (1) `envs/genesis.py`: shaping γ from a ctor kwarg (`pick_shaping_gamma`, was hard-coded 0.999) with φ(terminal)=0, plus `R2D_SIM_VARIANT` hook; (2) `envs/__init__.py`: passes `1 − 1/horizon` = 0.997; (3) `demo_prefill.py`: native stride-4 ingestion (`repeat.json` stamp must equal `action_repeat`, `demo_downsample=1`, terminal reward 1.0 × reward_scale); (4) `eval_genesis.py`: `--ic-file/--ic-set/--ic-index`, labelled outputs.

```
knob value pointer
---------------------------------------------- -----------------------------------------------------------
----------------------------------------------------------- --------------------------
----------------------------------------------------- --------------------------
Config configs/env/genesis_pick_v5d4c_delta_shaped.yaml (dense; cluster yaml; sbatch_r2dreamer.sh:190
sparse = same without _shaped)
Budget 3e6 sim steps (incl. prefill) = 750k decisions; 6 CPU worlds yaml :33-34
action_repeat / action space 4; delta_joint cap 0.025, leash mult 5; actor yaml :36,46-51; _base_.yaml:172-174;
METHODOLOGY.md §7.3
bounded_normal_clipped (min_std 0.1, max_std 1.0)
time_limit 400 sim steps (100 decisions) for every result cell yaml :40; sbatch_r2dreamer.sh:240;
CRITIQUE_decisions Q6
(TIME_LIMIT=400); the PREREG §8 pilot at 1200 failed (0/2
seeds at 2e6) → PREREG's stated on-fail action "keep 400" —
amendment not formally logged [CHECK 21]
Replay buffer.max_size 450000 rows, FIFO (LazyTensorStorage); sbatch_r2dreamer.sh:307; yaml :59-60; buffer.py:15;
demo_duplicate 4; demo_reinject_every 150000 online steps (→ AUDIT_results §1
19 re-injections/run; ring share ≈10 % / ≈27 % fails arm);
buffer not persisted across requeue
reward_scale 100 (env and demo) yaml :61
train_ratio 512; batch 16 × 64; pretrain 1000 yaml :39; configs.yaml:17-18,30
horizon → discount 333 → 0.997; imag_horizon 15; λ 0.95; return_clamp 100 yaml :65,82; _base_.yaml:17,19;
dreamer.py:494,502-503,549
actor_bc_lambda 0 (demos consumed as data only, by design) yaml :52
act_entropy 3e-5 yaml :66
Losses barlow 0.05, rew 1, con 1, dyn 1, rep 0.1, policy 1, value _base_.yaml:16,35-48,79
1, repval 0.3; kl_free 1.0; unimix 0.01
Model size12M: deter 2048, hidden 256, discrete 16, stoch 32, cnn configs/model/size12M.yaml;
depth 16, units 256, SiLU, norm; symexp-twohot 255-bin _base_.yaml:68-81,134-146,185-197; yaml :68-73
reward & critic heads; CNN encoder only (cnn_keys: image)
Optim lr 4e-5, AGC 0.3, warmup 1000, Adam (0.9, 0.999); slow _base_.yaml:25-33
target update every step, fraction 0.02
Demo sets baselines/matched_v2/r2d/<ARM> (dreamer-native, terminal sbatch_r2dreamer.sh:204-259
reward 1), stamp gate
Selection every latest.pt archived and scored in-job on sel (15 head -1 → BEST_selected.pt (byte copy,
confirmed); pruning sbatch_r2dreamer.sh:444-548
episodes in one process, --mode sample --seed 0 --device keeps best-2 + newest + K=5 fraction ckpts. Tie rule is
cpu) → ckpt_scores.tsv; `BEST = sort -rn sort`-order, not "later" [CHECK 22]
Protocol readout cluster/r2d_rescore.sh: BEST_selected.pt on hold and rnd, r2d_rescore.sh:1-53; commit bf09060
one fresh process per episode, CPU, PAR=4, --max-steps 1200,
asserted denominators, non-zero exit on missing → RESCORE-
RESULT (replaces an inline loop that printed DONE over 540
crashed processes)
Seeds (all old world, matched_v2, dense, TL 400) dR2D 80–93; dR2DDPfails 80–93 (s83 lost, A9); dH 99–108
registry; SESSION_LOG:167,173,179; N16
(s102 → s108); dDP 100–107; controls dR2Ddup13 200–203,
dR2DDPsucc 300–303; earlier 08-19 cells dH 50–53 (dense),
dR2D 40–43 (sparse); corrected-world pilot 60/61 (TL 1200)
Not run PREREG §1 sparse arm and 6 seeds per source at TL 1200 PREREG:35
(dense only, TL 400 ran) [CHECK 21]
Resources `-p gpu,preempt --requeue --gres=gpu:1 --constraint="l40s\ a100\ l40\ h200" -n 8 --mem 48g --time 2d
(relaunches with --time sbatch_r2dreamer.sh:162-177
24:00:00`); separate py3.11 venv
```

UPDATE 08-29 — what is standard. The replay-buffer critic loss (`loss_scales.repval 0.3`, `dreamer.py:539-563`) matches the official DreamerV3 v2 recipe (`beta_repval 0.3`, danijar/dreamerv3; `paper/WM_CANDIDATES_2026-08-29.md`); only the imagined-return clamp (`return_clamp 100`, `dreamer.py:502-503, 552-553`) is a deviation (E8 corrected). **Standard arm = `env.return_clamp=0`, repval kept** (runs `pick_v5d4c_delta_shaped_NOCLAMP_{dH,dDP}_s12{2,3}`, packed) — result 08-29 22:00: value runaway (train/val 115–825 vs max 100 at ~0.7M); the clamp is load-bearing. The earlier "STD" attempt (clamp off + repval off, s120/121) died at launch on a non-existent hydra key and is void. LAST-checkpoint hold re-scores (21 runs, `baselines/outputs/n12_rescore/*_LAST_hold_*`): 0/15 or ≥10/15 in 16/21 — final policies are bistable; BEST-of-K selection on `sel` (training ICs) does the work and must be disclosed as such. dup13 pilot (s202/203, sel-only): 0.93 / 0.00. Corrected-world gate pairs dH/dDP s80–83 (`DEMOSET=w3`, `gc_kp4_riser3_shelf6`, dense, 3M): **ignites** — sel max per seed dH 0.40/0.93/0.93/0.93, dDP 0.00/0.87/0.13/0.93, bistable across adjacent checkpoints as in the old world; protocol re-scores (BEST hold/rnd, LAST hold; tags `_W3`): dH BEST hold 10/15/11/10, rnd 20/25/22/16; dDP 1/2/2/15, 1/2/3/28 (n 4 v 4, direction human > DP, perm p 0.143); seeds 84–87 per source added 08-29 (4-seed packs, `-n 32`). Under A19 the WM arm reports in one named world per table; the w3 run dirs carry no world in their name (seeds 80–83 of dH/dDP = corrected world only). Packing: `cluster/sbatch_r2dreamer_pack.sh` (2 runs/GPU, 8 threads each, 16 CPUs, 96 GB; >2× per-GPU throughput, same launcher/config per run; 4/GPU with `-n 32` on 64-core nodes from 08-29). Touchgoal probe (no demos, `DEMO_DIR=null` now accepted by the launcher) s0/1 packed.

5.4 dv3 — DreamerV3 (dreamerv3-torch-genesis) — documented negative

Patch `cluster/patches/dreamerv3-torch-genesis_final_rr.patch`: converter refuses contract-v1 dirs / `--terminal-guard on`; `genesis_eval.py` gains `--ic-file/--ic-set/--ic-index/--device` and records `act_selection='deterministic'`; overlay `genesis_final_rr: time_limit 1200, steps 1e6`; new `sbatch_genesis_final_rr.sh` (K=5 archive, fresh-process eval sel → hold+rnd, `DV3-RESULT`). Config: action_repeat 4; time_limit 1200; steps 1e6 sim (250k decisions = 62.5k updates); train_ratio **256**; batch 16 × 64; prefill 2500 / pretrain 100; reward_scale 100; discount 0.997, λ 0.95, imag_horizon 15; RSSM deter 512, hidden 512, stoch 32×32, GRU blocks 8, units 512; model_lr 1e-4; dense variant `genesis_pick_shaping: True` γ 0.997 (`C/dreamerv3-torch/configs.yaml:40-58,94,101-104,384-492`). **Status: the final-RR dv3 block never ran** (registry: 0 rows of `sbatch_genesis_final_rr.sh`; one 2e4 smoke). The dv3 negative (N4: transient takeoffs at 140–320k, no sustained ignition) rests on the 08-19 `sbatch_genesis_multi.sh` 300k runs (dH s0–2, dDP s0–1, dR2D s0–1), whose harness is outside this repo **[CHECK 23]** (`PAPER_NOTES.md:87-92`; `CRITIQUE_launch_plan_2026-08-25.md:415`; `PREREG:37`). Earlier history: unnormalised demo actions poisoned runs v6–v13 (`METHODOLOGY.md §6.3`).

UPDATE 08-29 — no longer "documented negative"; mechanism + standard arm. Diagnosis (`paper/DV3_DIAGNOSIS_2026-08-28.md`): on every unclamped run `value_mean` runs past the attainable return (dense ×100: 263–390 by 300k on 3/3 5M baselines; touchgoal-from-scratch s0 learned to 0.82–0.96 success at 230–440k then collapsed as value crossed 100) — critic runaway via bootstrapped reward-head leak; the torch port (NM512, pre-v2) lacks the official replay critic loss / EMA regulariser. Arms (dH dense, seeds 20–22 at 1M; launcher `EXTRA_CONFIGS`/`TAGSUF`): **`genesis_dv3std`** = `genesis_reward_scale 1` (symlog/two-hot as published), `return_clamp 0`, `train_ratio 512`, `time_limit 400`, demos = `matched_v2/r2d/dH` (terminal 1.0, grant_slack 0) — the ONLY arm eligible for the matrix (A15); `genesis_dv3clamp` (clamp 100, diagnostic); `genesis_dv3fix` (clamp + train_ratio 512 + TL 400, diagnostic). std s20: first picks ever (train success 0.22 @284k) with value bounded (~29); clamp s20: 0.25 with value pinned at 98. 3M std pack (s23–25, `cluster/sbatch_dv3_pack.sh`, 3 runs/GPU, 375k updates ≈ r2dreamer's 373k) with an `afterany` resume chain (dreamer.py re-enters `latest.pt`). Disclosures: (i) every dv3 job before 08-28 21:30 ran WITHOUT its overlay (launcher omitted `${EXTRA_CONFIGS}` on the real train/eval lines) — the 5M baselines were unaffected; (ii) two other launcher defects (env python, relative demo dir) killed the 08-28 morning batch at start; (iii) `models.py` clamp is flag-gated (`return_clamp`, default 0). Eval uses `actor.sample()` labelled deterministic (mislabel, measurement only). Speed: ~47k sim steps/h (train_ratio 512) unpacked.

5.5 SACfD (legacy)

`baselines/rl/train_sacfd_full.py` — demo-seeded SB3 SAC (no PER, no n-step, FIFO eviction, staged reward, 400k steps). Superseded by RLPD; retained only as history (absolute-action fling artefact; hardened predicate) (`train_rlpd.py:1-22`; `AUDIT_sources_2026-08-23.md:374`; `METHODOLOGY.md §6.2`).

5.6 Cross-learner invariants

All learners: delta_joint, cap 0.025, leash 0.125, repeat 4, sparse terminal +1 (×100 for WMs), tip rule, same tapes (per world), same IC file, K=5 checkpoints, selection on `sel`, headline on `hold+rnd`, 1200-step eval horizon. Differences that remain (disclosure, PREREG §10): state vs pixel modality; per-learner budget and unit (DP 100k grad steps; RLPD 100k decisions; r2d 3e6 sim steps; dv3 1e6); r2d training horizon 400 vs 1200 for the others; r2d/dv3 have replay critics that regress through recorded fail states; recipes tuned on dH; RLPD has no passing positive control.

---

6. Evaluation protocol

6.1 Initial conditions — `baselines/eval_ics.json` (generated once by `baselines/make_eval_ics.py`, seed 0, git dd58c15)

```
set content pointer
--------- -----------------------------------------------------------
---------------------------------------------
sel (15) demo uids 232 234 235 236 237 239 242 243 244 245 246 247 eval_ics.json; make_eval_ics.py:46,91
248 250 251 (= the historic first-15 demo ICs) — selection
only
hold (15) 252 254 256 265 273 276 284 294 295 302 311 325 327 331 335, make_eval_ics.py:55-58,94-97; PREREG:277-
280;
drawn default_rng(0) from the 46 remaining success uids AUDIT_prelaunch:114-118
(61-uid universe), disjoint from sel — confirmation split of
demo ICs, not held-out (11/15 are training ICs of a 51-demo
set; for RLPD every sel/hold uid is a training reset IC; A8)
rnd (30) support ICs: can xy ~ Uniform over the solved-demo bounding ic_sampling.py:12-34; eval_ics.json
support_box
box ±1 cm, lo (0.3395, −0.2370), hi (0.592, 0.1887),
upright, z 0.113; fixed goal; the only novel ICs
```

```
Ceiling uid 234 (sel[1]) is a lying-can IC: picked 0/430 policy- PAPER_NOTES.md:250-255; AUDIT_INDEX.md:51
evals across 22 DP runs → sel max = 14/15 = 0.933; hold has
~1 count of headroom; BC source claims therefore read on rnd
(N11). N11's "tip rule fires at decision 1" wording
describes the training env, not the eval env (no tip
termination there) [CHECK 24]
```

6.2 Harness (`cluster/eval_sweep.sh` → `baselines/wandb_eval.py`; r2d via `eval_genesis.py`/`r2d_rescore.sh`)

- One episode per fresh process (`--ic-file --ic-set --ic-index k --seed k --no-wandb --max-steps 1200`), ≤1 world per 2 cores (`-n 8` → 4 concurrent), resume-safe; missing episodes reported as MISSING, never 0; denominators asserted from the IC file; `SWEEP-RESULT` line is authoritative (`eval_sweep.sh:37,60-66,114-166`).
- Sim always CPU (`GenesisCanEnv(backend='cpu', max_steps=1200)`); DP policy on GPU when visible, SAC on CPU (A4); sim_variant from CLI > sidecar > base, mismatch refuses; action mode/repeat/delta_ref from the checkpoint sidecar (no silent defaults) (`wandb_eval.py:122-152,171-210,281-326`).
- **Pick metric at eval** = the hardened predicate of §3.5 computed by `GenesisCanEnv` (no settle); episodes run the full 1200 steps (`done = t ≥ max_steps` only — **no tip termination at eval**); `placed`, `contact`, `nested` (100-step settle) also recorded but not used for pick-scope headlines (`genesis_can_env.py:255-294`; PREREG §10).
- Action selection: RLPD deterministic; DP sampled with per-episode seed; r2d sampled (`mode` reported alongside); dv3 deterministic — recorded in the result JSON (`act_selection`) with node class, git, sim_variant, ckpt step, n_steps, per-episode records (`wandb_eval.py:436-449`).
- r2d selection evals run 15 episodes in one process on `sel`; the protocol readout is the separate fresh-process re-score (§5.3).

6.3 Selection and headline rule (PREREG §5, A7)

K=5 checkpoints at 20/40/60/80/100 % of budget; selection = best `sel` (ties → later for DP/RLPD); **headline per seed = the selected checkpoint on `hold` (15) + `rnd` (30)**, reported by name ("sel / hold / rnd"), plus final-checkpoint hold/rnd and time-to-first-training-pick. Results are quoted only from `analysis/results_table.py` output (`SWEEP-HEADLINE`, `HEADLINE.txt`, `RESCORE-RESULT`; `R2D-RESULT` rows are flagged sel-only), deduplicated on (arm, seed), smoke seed 0 excluded, world inferred from wave/seed block (`results_table.py:10-40,64-74`).

6.4 Seeds per cell (registry mirror as of 08-28; `RESULTS_TABLE_2026-08-25.md` shows the older n=4/5 snapshot [CHECK 25])

```
learner world reward arms n
-------------------- --------- ------------ -------------------------------------------
----------------------------------------------------------
DP old – dH, dDP, dR2D 5 each (10–14)
DP corrected – dH, dDP 10 each (20–29)
DP corrected – dHallpruned_1e3/1e2, dDPallpruned_1e3/1e2 (N7) 3 planned (1–3 landed)
DP corrected – dH_A/B, dDP_A/B (N14) 3 planned (2/12 landed at audit time; resubmitted 08-28)
RLPD old sparse dH, dDP, dR2D, dDPfails, dR2DDPfails 4 each (10–13)
RLPD old dense dH, dDP, dR2D 4 each
RLPD corrected sparse, dense dH, dDP 10 each (20–29)
r2dreamer (RESCORE) old dense dR2D / dR2DDPfails 4 / 3 at audit (80–83 / 80–82; more re-scores in flight; s83
```

```
fails lost)
r2dreamer (sel-only) old dense dH, dDP 3 / 4 (not reportable)
dv3 — — — 0 at the final-RR budget
```

6.5 Statistics (PREREG §7 as amended by the 08-28 audit)

- Unit = seed; primary statistic = per-seed success count on `hold` and on `rnd` of the selected checkpoint; paired by seed index within a learner; no pooling of episodes across seeds (episode-pooled Fisher tests are invalid and must not be quoted).
- Tests: Welch t on seed means with **Welch–Satterthwaite df** (N15: hold df 4.52 → CI [−0.259, +0.726]; rnd df 4.17 → [−0.228, +0.556]); exact two-sided permutation test (N15 p = 0.286; **minimum attainable p at 4 v 3 = 2/35 = 0.057**, at 3 v 3 = 2/20 = 0.100); 95 % bootstrap CI on the mean difference as effect size; hierarchical Beta–Binomial P(A>B) as the reported posterior; two primary contrasts per learner → Holm; everything else BH q = 0.1, labelled exploratory (`PREREG:172-175,194-201`; `AUDIT_results §1`).
- Run-identity rule A9: a seed id names one run; a rerun replaces; earlier readouts dropped and listed as lost.
- **[CHECK 26]** the only statistics code in `analysis/` is `bayes_source_effect.py` (08-08 wave, wandb `eval_indist/eval_random` keys) and `results_table.py` (means only); the Welch/permutation/Holm computations quoted in the notes were done ad hoc and are not scripted.

UPDATE 08-29. `analysis/stats.py` (Welch with Welch–Satterthwaite df and the correct critical value, exact permutation p under both conventions, minimum attainable p; self-test reproduces the corrected N15 CIs) resolves **[CHECK 26]**. RLPD statistic per A16; WM tables per A19 (one world per table); fails effect per A17 reading rule.

7. Compute, provenance, incidents

```
item value pointer
--------------------- ------------------------------------------------------------
------------------------------------------------------
Cluster Tufts HPC (Slurm); GPUs L40S / A100 / L40 / H200; one GPU sbatch_*.sh headers;
sbatch_r2dreamer_pack.sh:5-6;
per training run; Genesis worlds CPU-side; partitions SESSION_LOG:165,173,178
gpu,preempt --requeue for training, gpu for relaunches,
batch for CPU re-scores (-c 8 --mem 24g --time 4h); per-user
QOS cap 10 GPUs; long jobs submitted with --time 24:00:00
(no documented 24 h hard cap [CHECK 27])
Environments one pip-only conda env C/condaenv/genesis for genesis + verify_env.sh:24-58; install_r2dreamer.sh
lerobot + SB3 (verified by cluster/verify_env.sh, incl.
H200); r2dreamer venv C/r2d_venv (py3.11, torch 2.8)
Dev box Ryzen 5950X / RTX 3080Ti / 62 GB; pilots and labs only; METHODOLOGY.md:380-383
official numbers never mixed across machines
Run registry cluster/run_registry.py → cluster/RUN_REGISTRY.jsonl run_registry.py:12-18,57-67,92-102,156-216;
(cluster copy authoritative, 245 rows 08-18→08-28 in the sbatch_rlpd.sh:203-235; sbatch_dp.sh:352-381
mirror; local file is empty [CHECK 28]). Row fields: script,
arm, seed, knobs, demo_fingerprint (sha of sorted
(filename,size)), git, semantic_key, full_key, timestamp,
slurm_job_id, duplicate_ok_reason, stage, node; registered
at job START; full-key duplicate → REGISTRY-REFUSE unless
DUPLICATE_OK=<reason>; semantic duplicate → REGISTRY-WARN
(no warn field written yet, contrary to A9 [CHECK 29])
Registry knobs (RLPD) steps, budget_unit, scope, action_mode, delta_ref, sbatch_rlpd.sh:203-208
action_repeat, train/eval horizon, gamma, backup_entropy,
```

```
per_member_ln, pick_hold_reward, utd, ensemble_size,
subset_size, demo_batch, reward, pick_shaping, demo_format,
demo_sha, wave, demo_shaping, terminal_zero, terminal_guard,
sim_variant
Session command log paper/SESSION_LOG_2026-08-23_cluster.md: every SESSION_LOG:42-43
create/run/submit/cancel/delete cluster command with EDT
time, host, repo git sha at run time, exact command, exit
code, first output line (146 rows + 21 NOTE bullets)
Artifact mirror ~/workspace/final_rr_artifacts_2026-08-24/ (9.7 GB): demo ls M/; AUDIT_INDEX.md §6
recordings + manifests, matched_v2 (incl. r2d/, dR2DDPsucc),
matched_w3, DP selected checkpoints and sidecars, RLPD
checkpoints (+meta), all .out files (696), HEADLINE.txt
(60), sweep.json (577), r2d re-score logs, registry,
pull.log
Committed provenance paper/final_rr_sets_2026-08-23/ (v1/v2 manifests +
censuses); cluster/patches/ (genesis pin + patch, r2dreamer
and dv3 patches); cluster/RUN_REGISTRY.jsonl (should mirror
the cluster copy)
wandb entity jambotime; projects genesis_paper (DP/RLPD train + eval_sweep.sh:171; sbatch_dp.sh:189;
sbatch_rlpd.sh:200;
sweep summaries), r2dreamer_genesis, dreamer_v3, memory monitoring-box-setup.md
genesis_pickaplace (wandb_eval default)
Code identity PREREG §2 "one commit hash per block; a fix restarts that SESSION_LOG sha column
learner's block" — the block spanned several shas (e.g. v2
sets ca05472, w3 sets 2061f08, later relaunches
0440a7f/bf09060); every registry row and manifest carries
its sha, so per-run identity is exact even though the block-
level rule was not kept [CHECK 30]
```

Incidents affecting data (all logged; report in the paper's limitations):

1. **Disk incident (08-27)**: K=5 archiving shipped with no retention → `baselines/outputs` = 909 GB, shared volume 100 %; jobs died with bare exit codes (the same signature previously attributed to preemption/CUDA transients — that diagnosis is partly invalid); 176 checkpoint dirs pruned keeping selected + final of every completed run; 489 GB recovered; of the 20 DP resubmits of 08-27 02:13, 8 completed and 12 failed on the full volume; N14 had 2/12 and N7 9/12 landed before resubmission 08-28 (`SESSION_LOG` NOTE 08-27; `AUDIT_results §0,§6`; commit `30195ee`).
2. **Seed overwrite (A9)**: the 08-27 20:15 relaunch reused seed ids of completed runs — dR2DDPfails **s83** (sel 0.80, BEST written) overwritten in place and unrecoverable; dH s102 in-place relaunch cancelled 08-28 04:30 and resubmitted as **s108**; dDP s101 relaunch never ran. Pooled statistics drop the lost readouts (`PREREG A9`; `SESSION_LOG:178-179`).
3. **Over-broad `scancel --name=r2d-train --state=PENDING` (08-26)** killed the N13 and N12 top-up waves; 11 relaunched with `DUPLICATE_OK=relaunch-after-overbroad-scancel` (`SESSION_LOG:162-164`).
4. **Re-score false success**: an inline re-score loop printed DONE over 540 crashed processes (no exit-code check; missing `GENESIS_PICKAPLACE_ROOT`); replaced by `cluster/r2d_rescore.sh` with asserted denominators (`r2d_rescore.sh:11-15`; commit `bf09060`).
5. **E3 exposure arm cancelled** (BUFFER_MAX=40000 confounded staleness/capacity/reinject shock; 10.1 % vs 34.2 % demo share) — replaced by the N16 controls (`AUDIT_INDEX.md §4`).
6. **Audit-day duplicate submissions**: 24 DP jobs submitted with a wrong guard path, 11 duplicates cancelled before start (`AUDIT_results §6`).
7. Earlier instrument failures that shaped the protocol (fling exploit, env-vs-replay #26 horizon bug, unnormalised dv3 demos, eval action-mode default, six silent-default bugs): `paper/METHODOLOGY.md §8.4`; `CLAUDE.md`.

UPDATE 08-29. GPU packing (CPU-bound Genesis, 0 % GPU util): r2dreamer 2–4 runs/GPU, dv3 3 runs/GPU, RLPD unpacked (GPU-bound at UTD 10). QOS cap 10 GPUs; `gpu` MaxTime 2 d; `scontrol top` denied → queue order via `scontrol hold` + `--begin` release jobs. Unattended readouts: `cluster/harvest_readout.sh` on the batch partition at +3/+20/+44/+62 h → `paper/harvest_<ts>.md`. Incidents 08-28/29: three silent pack deaths (40 s, found ~8 h later) → rule: check every pack `.out` for `rc=1` within the first hour. Fallback WM (A18): DEMO3 codebase (TD-MPC2-based; MoDem / TD-MPC2 by flags) prepared locally (`~/workspace/demo3_prep`, converter round-trip verified); `policy_pretraining=false` flag set is actor-BC-free.

8. [CHECK] list (every uncertain or conflicting value)

1. **Recording rate**: "30 Hz" (docs) vs ~31.6 Hz (uid 232 `_episodes.npy`, 919 frames/29.06 s) vs ~40 Hz `joint_states` / 1159 frames in `_cartesian.npy` for the same trial; timestamps not saved (`example.py:276`; `CAN_STARTING_POSITION.md:188`).
2. **`vel_cmd` semantics**: the stored arm stream is measured `/joint_states` position, not a velocity or a command; several docs say "commanded joint targets". Fix wording throughout (`trial_reader.py:69-74`; `METHODOLOGY.md:233-235,281-284`).
3. **Real camera resolution/model**: not documented anywhere.
4. **93 vs 96 trials**: uids 253, 285, 289 (fail list) on disk but absent from `trial_placements.json`/manifests, reason undocumented; also `kinova.py` lists 322 and 331 as both success and fail; "All 224 bags are local" (`HANDOFF_PLAN.md:87`) vs 96 processed.
5. "~130 recordings" for unlabeled uids 200–231 (`HANDOFF_PLAN.md:91`) — inconsistent with a 32-wide range.
6. **Contact-solver default**: `replay_harness.py:98-99` comment claims 0.2.1 hard-wires timeconst = 2·substep_dt; gripper lab shows this tree keeps 0.02 s and only clamps. Confirm which statement holds for the exact pinned commit 31951c3f.
7. **Finger kp**: `CAN_STARTING_POSITION.md:172-174` says kp 100 "sweet spot"; live world config is `finger_kp 40` (commit e8a07d6).
8. `sim_variants.post_build` does not assert the 5-geom count for `grasp_timeconst` (silent degrade possible while stamping `_ts5`).
9. **Human pick-recreation figures** come from different runs: 51/66 (meas, cluster), 56/66 (either, cluster, base), 57/66 (gripper-lab g_base control) vs 58/66 (ts5, cluster-confirmed), 58/66 (dH_w2 in gc_kp4_riser3_shelf6), 55/66 and 58–62/66 (local follower-lab variants). State which comparison each number belongs to.
10. **ts5 / `matched_w4` fate**: `dH_w4` recorded 08-26 07:50 (ts5) and `matched_w4_pilot/dH` built, but no kept count, census, or DP-pilot readout appears in any note or results table; `matched_w4` proper never built; commit 7c8195d calls ts5 "world of record from 08-26" while all subsequent result launches used `gc_kp4_riser3_shelf6`. The draft states "ts5 = pilot only, no result block".

11. `genesis_can_env.py:10` docstring says 16-dim state; code is 17.
12. **Tip penalty**: code `TIP_PENALTY = 0.0` (pick/full scope) and PREREG §2 r=0; the task brief's "−0.5 in training" is the July Cartesian env only.
13. **Fast-pick 2.0 mechanism**: the "picked + placed grants in the same window" explanation is read from `genesis_can_env.py:265-267`; the only written source is a comment in `to_dreamer_native.py:65-70` ("stage grant AND the hardened-pick terminal inside ONE repeat-4 window").
14. `dHunpruned` control was recorded twice but no result for it exists in any results file (PREREG §1 lists `dHunpruned_DP ×3`).
15. PREREG A1's drop taxonomy ("5 over-horizon, 7 over-cap") vs recorder outcome codes (10 env_truncated, 2 adapter_exhausted, 3 tipped) for the same 15 drops.
16. `sbatch_dp.sh:14,221` says lerobot datasets are "prebuilt by make_matched_sets.py"; they were built by `convert_to_lerobot.py` in separate steps.
17. Split-half set shas not captured locally.
18. The "22 %" pruned fraction at eps 1e-2 exists in no artifact.
19. DP 300k "epoch-matched view" (PREREG §1) never ran.
20. RLPD old-world seeds are 10–13 (n=4), not 10–14; DP old-world seeds are 10–14 (n=5).
21. r2dreamer ran dense only, TL 400, old world — PREREG §1 planned sparse+dense × 6 seeds at TL 1200 (pilot-gated); PREREG §8's on-fail action ("keep 400") was taken but no amendment was logged (CRITIQUE Q6).
22. r2dreamer checkpoint tie-break is `sort -rn | head -1` (not "later"); its in-job selection evals are 15 episodes in one process (not fresh-process).
23. dv3 final-RR block never ran; the N4 negative rests on 08-19 300k runs whose harness (`sbatch_genesis_multi.sh`) lives outside this repo; `results_table.py` has no DV3 parser.
24. N11's "tip rule fires at decision 1" mechanism for uid 234: the eval env has no tip termination; the 0/430 is real but the wording describes the training env.
25. `RESULTS_TABLE_2026-08-25.md` shows n=4/5 for corrected-world cells; the mirror/audit have n=10. Regenerate before quoting.
26. No scripted Welch/permutation/Holm/BH analysis exists in `analysis/`; `bayes_source_effect.py`/`results_significance.md` describe the 08-08 wave.
27. No document states a 24 h cluster wall-clock cap; only `--time 24:00:00` usage is observable (script defaults 12 h / 14 h / 2 d).
28. Local `cluster/RUN_REGISTRY.jsonl` is 0 lines; the mirror copy (245 rows) predates seeds 104–108, 200–203, 300–303.
29. Registry rows carry no `warn` field (A9 requires one); 0/245 rows have it.
30. PREREG §2 "one commit hash per block" was not kept (several shas within each block); per-run shas are exact.
31. E5: the ts5 change also stiffens can↔goal-can and can↔shelf contacts 1.6× — relevant only if any downstream (contact/nested) number from the gripper lab is quoted.
32. `GenesisCanEnv.placed` still tests the un-shifted `BOX_TOP_Z = 0.11` under shelf6 worlds (a correctly placed can scores False); inert for pick scope

(`CRITIQUE_decisions:455-461`).